

Objective

To implement and demonstrate Multiple Inheritance using Interfaces in Java, where the Child class implements both Father and Mother interfaces.

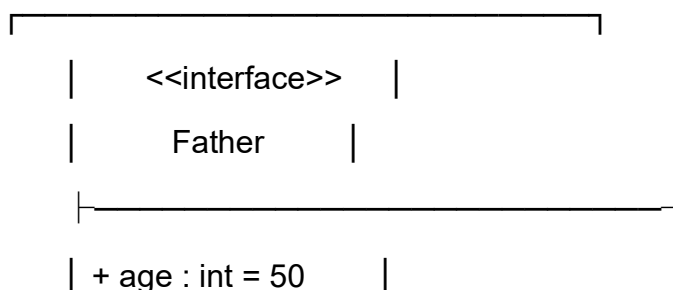
Steps for Execution

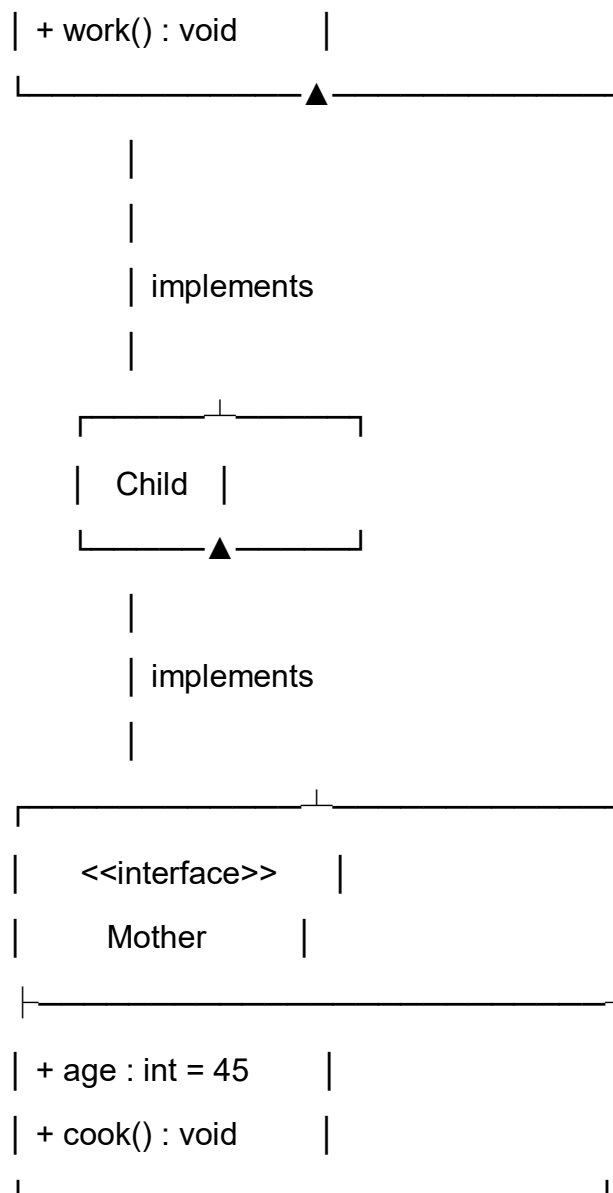
1. Create a Java project and package named javacore.
2. Create a file named Child.java and enter the given code.
3. Define the Father and Mother interfaces.
4. Implement both interfaces in the Child class.
5. Compile and run the program.
6. Observe the output.

Algorithm

1. Start the program.
2. Define the Father interface with age and work().
3. Define the Mother interface with age and cook().
4. Create the Child class implementing both interfaces.
5. Implement the work() and cook() methods.
6. Create a Child object.
7. Call work() and cook().
8. Display the output.
9. Stop the program.

UML CLASS DIAGRAM:





Relationships

- Child implements Father
- Child implements Mother
- Father and Mother are interfaces.
- Child provides implementations for `work()` and `cook()`.

CODE:

```
package javacore;
```

```
interface Father {
```

```
    int age = 50;
    void work();
}
```

```
interface Mother {
    int age = 45;
    void cook();
}
```

```
public class Child implements Father, Mother {
    public void work() {
        System.out.println("Father works");
        System.out.println("Father age: " + Father.age);
    }
    public void cook() {
        System.out.println("Mother cooks");
        System.out.println("Mother age: " + Mother.age);
    }
    public static void main(String[] args) {
        Child c = new Child();
        c.work();
        c.cook();
    }
}
```

OUTPUT:

Father works

Father age: 50

Mother cooks

Mother age: 45

Result

The program successfully demonstrates Multiple Inheritance using Interfaces in Java. The Child class implements both Father and Mother interfaces and accesses their respective methods and variables.